MTH101, Basic Linear Algebra

Monsoon 2025

25/06/2025, 2:00PM-3:00PM.

Tutorial-4

(1) Let J be an $n \times n$ matrix in Echelon form. Verify that either J has a zero row or J is the identity matrix.

(2) Let E_{ij} be the $n \times n$ square matrix such that $a_{ij} = 1$ and 0 everywhere else. Verify that $E_{ij}E_{jk} = E_{ik}$ and that $E_{ij}E_{pq} = 0$ if $j \neq p$.

(3) Show that elementary matrices[1] are invertible by verifying that
   (i) The inverse of the "row swap matrix" $B = I_n - E_{ii} - E_{jj} + E_{ij} + E_{ji}$ is B itself.
   (ii) The inverse of the "row replacement matrix"$I_n + \alpha E_{ij}$ is $I_n - \alpha E_{ij}$.
   (iii) The inverse of "scaling matrix"$I_n + (\alpha - 1)E_{ii}$, where $\alpha \neq 0$, is $I_n + (\frac{1}{\alpha} - 1)E_{ii}$

(4) If $n \times n$ matrices A and B are invertible then their product AB is also invertible.

(5) If the row reduced Echelon form J of a matrix A is the identity matrix then A is invertible[2].

[1] matrices that represents row operations

[2] partly done in class on 24 September 2025. Use the problems 3 and 4